

|                  |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                      |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
|                  | hence, the mean e.m.f. induced in coil P is <u>larger</u> .                                                                                                                                                                                                                                                                                           |                                                                                                                      |
| <b>6(a)(i)</b>   | <p>Wave nature contradicts: Since energy is arriving in <u>continuous manner</u>, a certain time is needed for the electron to gather enough energy before it is ejected.</p> <p>Particulate nature supports: Energy of the incident photon will be <u>instantaneously transferred</u> to the absorbing electron in <u>a one-one interaction</u>.</p> | <p>[1]</p> <p>[1]</p>                                                                                                |
| <b>6(a)(ii)</b>  | Work function of a surface is the <u>minimum</u> energy required to liberate an electron from the surface.                                                                                                                                                                                                                                            | [1]                                                                                                                  |
| <b>6(a)(iii)</b> | <p>By Conservation of energy,<br/>Kinetic energy of electron = Energy of photon – Work done to emit an electron from the surface.</p> <p>But electrons can be ejected from <u>different layers of metal</u>, hence work done can vary, resulting in different KE. When the work done is minimum, KE will be maximum.</p>                              | <p>[1] using conservation of energy or Einstein's photoelectric equation</p> <p>[1] for any correct explanation.</p> |
| <b>6(b)</b>      | <p>Energy of photon, 350 nm</p> $= \frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{350 \times 10^{-9}}$ $= 5.6829 \times 10^{-19} \text{ J}$ $= 3.55 \text{ eV}$                                                                                                                                                           | [2] for any correct calculations                                                                                     |

| No.     | Solution                                                                                                                                                                                                                                                                                                                                                  | Remarks                           |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
|         | <p>Energy of photon, 700 nm</p> $= \frac{hc}{\lambda}$ $= \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{700 \times 10^{-9}}$ $= 2.8414 \times 10^{-19} \text{ J}$ $= 1.78 \text{ eV}$ <p>To emit an electron, energy of photon must be greater than work function.<br/>So electrons will be emitted from potassium metal using light of 350 nm.</p> | <p>[1] for correct conclusion</p> |
| 7(a)(i) | Energy density is the amount of energy that can be released                                                                                                                                                                                                                                                                                               | [1] for correct                   |